Therefore,

$$(y+6)(y+8) = y^2 + 6y + 8y + 48$$

$$(y+6)(y+8) = y^2 + (6+8)y + 48$$

$$= y^2 + 14y + 48$$

(iii) $43 \times 36 = (40+3) \times (40-4)$

We know the identity

$$(x+a)(x+b) = x^2 + x(a+b) + ab$$

Taking, $x = 40$, $a=3$ and $b= -4$, we get

$$(40+3)(40-4) = 40^2 + 40(3-4) + 3(-4)$$

$$= 1600 + 40(-1) - 12$$

$$= 1600 - 40 - 12$$

$$= 1600 - 52$$

Therefore, $43 \times 36 = 1548$.

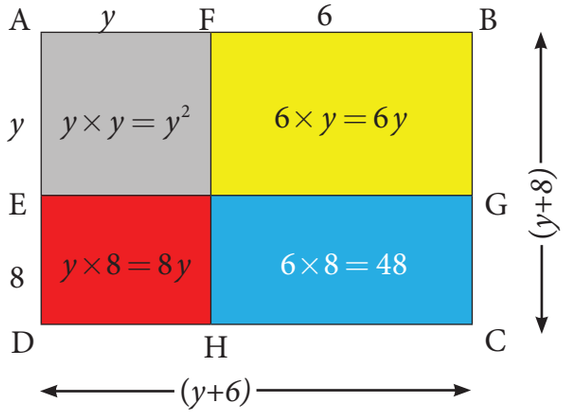


Fig. 3.12

3.3.2 Identity-2: $(a+b)^2 = a^2 + 2ab + b^2$

Consider four regions. Two square shaped regions with the dimensions of 3 × 3 (Grey) and 2 × 2 (Blue). Also, there are two rectangle shaped regions and both are in the dimension of 3 × 2 (yellow).

Arrange them in a square shape as shown in the *Fig. 3.13*.

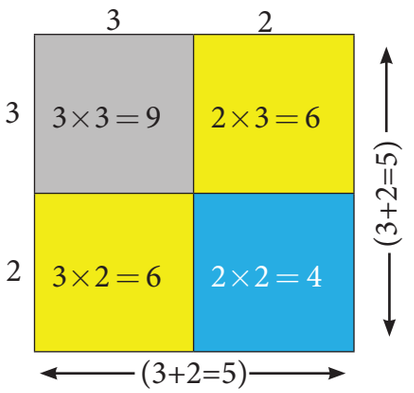


Fig. 3.13

By observing the *Fig. 3.13*, we can note that

Area of the bigger square = Area of two small squares + Area of two rectangles.

$(3+2)^2 = 3^2 + (2 \times 3) + (3 \times 2) + 2^2 = 3^2 + (3 \times 2) + (3 \times 2) + 2^2$ [since, 2×3 = 3×2]

Therefore, $(3+2)^2 = 3^2 + 2 \times (3 \times 2) + 2^2$

where, L.H.S is $(3+2)^2 = 5^2 = 25$ ...(1)

R.H.S is $3^2 + 2 \times (3 \times 2) + 2^2 = 9 + 12 + 4 = 25$. ...(2)

From (1) and (2), L.H.S = R.H.S.

Now, we can prove this for the variables a and b.

Let us take a square ABCD of side $(a+b)$, hence its area is $(a+b)(a+b) = (a+b)^2$

By observing the *Fig. 3.14*,

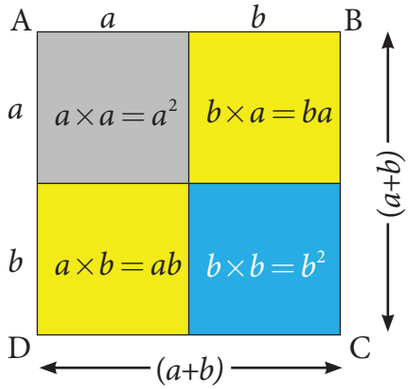


Fig. 3.14

Area of the bigger square ABCD = Area of two small squares (Grey and blue) + Area of two rectangles (yellow).

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